The Moving Essential Boundary of an Orbit-Resolved Hénon Reflection Euler Product

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Abstract

We construct the full Euler product carrying the individual symmetry-defect weights of every primitive odd reflection word in the full Hénon horseshoe. If A_n is the primitive marked packet and $S_n\chi$ is the orbit sum of $\chi(s) = \mathbf{1}\{s_{-1} = s_1\}$, set

$$\mathcal{Z}_{\mathrm{orb}}(z, q) = \prod_{n \text{ odd}} \prod_{\omega \in A_n} (1 - z^n q^{S_n\chi(\omega)})^{-1}.$$

Its logarithmic derivative has the exact primitive/repetition coefficient $\sum_{n|m} nE_n(q^{m/n})$, where E_n is the primitive moment polynomial. For every $q > 0$, the convergence radius is $R(q) = (1 + q^2)^{-1/2}$ and

$$\log \mathcal{Z}_{\mathrm{orb}}(z, q) = \frac{q}{\sqrt{1 + q^2}\,[1 - \sqrt{1 + q^2}\, z]} + G_q(z)$$

near the positive boundary, with G_q analytic. Thus the boundary is exponentially essential. The aggregate-mean radius is $(2q)^{-1/2}$ and is strictly too large unless $q = 1$. This closes the orbit-resolved analytic germ but leaves relative Lind/Fredholm renormalization and arithmetic semantics open.

1 From moments to individual Euler factors

For $H(x, y) = (6 - x^2 - y, x)$, the inherited full-horseshoe coding and reversor identify odd reflection packets with reversal-fixed binary words [2, 1]. Let A_n be the primitive marked packet and put

$$\chi(s) = \mathbf{1}\{s_{-1} = s_1\}, \qquad S_n\chi(s) = \sum_{j=0}^{n-1} \chi(\sigma^j s).$$

The primitive moment polynomial

$$E_n(q) = \sum_{\omega \in A_n} q^{S_n\chi(\omega)} \tag{1}$$

was computed exactly from the reflected rank-two transfer matrix:

$$F_{2m+1}(q) = 2q(1 + q^2)^m, \tag{2}$$

$$E_n(q) = \sum_{k|n} \mu(k) F_{n/k}(q^k). \tag{3}$$

For $q > 0$, define the orbit-resolved product

$$\mathcal{Z}_{\mathrm{orb}}(z, q) = \prod_{\substack{n\geq 1 \\ n \text{ odd}}} \prod_{\omega\in A_n} \left(1 - z^n q^{S_n\chi(\omega)}\right)^{-1}. \tag{4}$$

Unlike an aggregate-mean product, (4) retains every individual energy in (1).

Theorem 1.1 (Primitive and repeated weights)**.** *In the disk of absolute convergence,*

$$\log \mathcal{Z}_{\mathrm{orb}}(z, q) = \sum_{r\geq 1} \frac{1}{r} E(z^r, q^r), \qquad E(z, q) := \sum_{n \text{ odd}} E_n(q) z^n, \tag{5}$$

$$[z^m]\, z\partial_z \log \mathcal{Z}_{\mathrm{orb}}(z, q) = \sum_{\substack{n|m \\ n \text{ odd}}} nE_n(q^{m/n}). \tag{6}$$

Proof. Expand each factor of (4) by $-\log(1 - w) = \sum_{r\geq 1} w^r/r$. An r-fold repetition replaces both z^n by z^{nr} and $q^{S_n\chi}$ by $q^{rS_n\chi}$. Summing first over ω gives $E_n(q^r)$, proving (5); differentiation and collection of $m = nr$ gives (6). ☐

2 The two-variable primitive series

Proposition 2.1. *For $q > 0$ and sufficiently small z,*

$$E(z, q) = \sum_{\substack{k\geq 1 \\ k \text{ odd}}} \mu(k) \frac{2(qz)^k}{1 - (1 + q^{2k})z^{2k}}. \tag{7}$$

Its first positive singularity is

$$R(q) = \frac{1}{\sqrt{1 + q^2}}. \tag{8}$$

Near $R(q)$,

$$E(z, q) = \frac{q/\sqrt{1 + q^2}}{1 - \sqrt{1 + q^2}\, z} + G_{1,q}(z), \tag{9}$$

where $G_{1,q}$ is analytic.

Proof. Insert (2) into (3), multiply by z^n, and interchange the sums to obtain (7). The $k = 1$ denominator gives (8) and the displayed principal part. For odd $k \geq 3$,

$$(1 + q^{2k})^{1/k} < 1 + q^2$$

by the strict binomial inequality. Therefore their positive singular radii are all larger than $R(q)$. ☐

3 Moving essential boundary

Theorem 3.1 (Essential-boundary curve)**.** *For every fixed $q > 0$, there is a function G_q, analytic near $R(q)$, such that*

$$\log \mathcal{Z}_{\mathrm{orb}}(z, q) = \frac{q}{\sqrt{1 + q^2}\,[1 - \sqrt{1 + q^2}\, z]} + G_q(z). \tag{10}$$

Consequently $\mathcal{Z}_{\mathrm{orb}}(\,\cdot\,, q)$ has an essential singularity at $R(q)$.

Proof. The $r = 1$ term of (5) is handled by proposition 2.1. For $r \geq 2$, the first positive singular radius of $E(z^r, q^r)$ is

$$(1 + q^{2r})^{-1/(2r)} > R(q),$$

again by the strict binomial inequality. Hence all repetition terms are analytic near $R(q)$. Equation (10) follows, and exponentiating a simple pole gives an essential singularity. □

The object in (4) is still a restricted marked-packet product, not the full Lind zeta of the infinite-dihedral action [3, 4]. The latter includes all finite-index subgroup fixed-point data.

4 The exact cost of mean-field weighting

Write $q = e^{-s}$. The orbit-resolved pressure is $\frac{1}{2}\log(1 + q^2)$, while the aggregate-mean pressure is $\frac{1}{2}\log(2q)$. Thus their radii obey

$$\frac{R(q)}{R_{\mathrm{mf}}(q)} = \sqrt{\frac{2q}{1+q^2}} = \frac{1}{\sqrt{(q+q^{-1})/2}} \leq 1. \tag{11}$$

Equality holds only for $q = 1$. Mean-field weighting therefore predicts a strictly too-large disk at every nontrivial positive weight. At $q = 1$, (4) specializes coefficientwise to the unweighted canonical packet product.

5 Certificate and route boundary

The exact certificate constructs (6) as integer polynomials through degree 31, checks the rational values $q = 1/2, 1, 2$, and locks the $q = 1$ coefficients to the earlier product. An independent program audits (8)–(11) at five positive weights. Eight tests pass in normal and optimized modes, six dependencies are locked, and 24 hostile mutations are rejected.

The result closes a full orbit-resolved Route-A Euler germ and its boundary curve. Its essential singularity prevents naive Fredholm promotion but does not exclude a separately proved relative counterterm. No rational-prime labels, von Mangoldt amplitudes, Hilbert space, or self-adjoint operator have been constructed. Route B is not authorized. The next major problem is a source-native comparison with the full flip/Lind zeta and an explicit relative renormalization test.

References

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